

SKAGIT, WA, USA

WMO: 720272

Lat: 48.467N

Lon: 122.417W

Elev: 140

StdP: 14.62

Time Zone: -8.00 (NAP)

Period: 04-19

WBAN: 94282

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	24.9	27.9	12.1	10.3	27.1	18.7	14.1	30.9	20.9	48.6	19.0	48.7	4.2	40	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.1	81.3	66.0	78.5	64.5	74.8	63.3	67.3	76.7	65.9	75.3	64.5	72.6	7.9	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.0	89.6	68.8	63.1	86.8	67.9	61.3	81.4	66.9	31.7	76.6	30.6	75.6	29.6	72.5	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.7	15.8	13.0	DB	18.6	88.6	3.9	10.9	15.8	96.4	13.5	102.8	11.4	108.9	8.6	116.8	(4)
(5)			WB	17.4	70.4	4.2	2.4	14.4	72.1	12.0	73.5	9.6	74.8	6.6	76.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	41.3	42.3	45.7	49.1	54.9	58.7	63.0	63.5	59.4	51.2	45.2	40.4
	DBStd	9.54	6.47	5.43	4.88	4.36	4.28	4.03	3.80	3.48	5.33	4.45	6.14	6.14
	HDD50	1227	278	220	150	67	7	0	0	0	0	39	165	301
	HDD65	5083	736	637	598	478	315	195	86	71	182	429	595	761
	CDD50	1685	7	3	17	39	158	261	402	418	281	74	20	5
	CDD65	67	0	0	0	0	1	6	23	24	13	0	0	0
	CDH74	847	0	0	0	2	20	67	289	296	173	0	0	0
	CDH80	187	0	0	0	0	1	6	53	40	87	0	0	0
Wind	WSAvg	5.6	6.2	6.1	7.0	6.1	5.1	5.0	5.0	4.7	4.5	5.1	6.1	5.8
Precipitation	PrecAvg	43.5	5.3	3.4	4.1	3.5	3.1	2.5	1.1	1.4	2.2	4.4	6.7	5.6
	PrecMax	56.2	9.0	6.2	6.3	5.2	5.7	4.6	3.1	4.6	4.8	8.4	12.9	8.3
	PrecMin	33.7	2.9	0.5	1.5	1.0	1.1	0.5	0.1	0.1	0.7	1.3	2.5	2.9
	PrecStd	5.9	1.5	1.4	1.2	0.9	1.2	1.2	0.7	1.2	1.3	2.0	2.6	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	57.3	63.9	69.8	77.0	80.7	84.4	83.9	103.6	67.7	63.7	58.9
		MCWB	53.1	49.1	53.2	58.6	62.8	65.8	67.4	66.7	52.7	58.5	57.1	54.7
	2%	DB	55.0	54.8	59.2	63.8	72.2	75.1	81.2	80.9	77.4	63.8	58.9	54.4
		MCWB	51.1	49.3	50.9	54.9	60.6	63.2	66.0	65.9	63.4	56.9	54.4	51.3
	5%	DB	53.7	52.3	55.2	60.8	67.9	72.2	77.1	77.2	72.5	61.4	55.4	52.0
		MCWB	50.8	48.2	49.7	52.4	58.8	62.0	63.8	64.0	61.9	55.8	52.2	49.1
	10%	DB	50.3	50.0	54.1	56.8	64.1	68.5	73.3	73.5	69.5	59.0	54.1	48.4
		MCWB	47.6	47.2	49.5	49.9	56.7	60.0	62.9	63.1	61.1	54.2	51.4	46.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.8	53.0	55.3	59.7	63.6	68.1	70.6	69.1	67.4	62.2	58.9	56.2
		MCDB	57.8	54.7	59.7	66.6	72.8	75.6	80.3	78.7	73.9	64.3	61.8	58.2
	2%	WB	53.6	50.4	53.6	56.3	61.6	65.1	67.2	67.1	65.3	59.5	55.3	52.4
		MCDB	54.9	52.6	57.1	62.3	70.0	73.0	77.1	77.5	73.0	62.6	58.3	54.0
	5%	WB	50.5	48.5	51.6	53.8	59.6	62.9	65.7	65.8	63.2	57.1	53.6	49.0
		MCDB	52.2	50.7	55.2	58.4	66.3	69.2	75.5	75.6	68.0	60.3	55.9	50.9
	10%	WB	47.8	46.9	49.6	51.6	57.6	61.2	64.2	64.5	61.6	55.2	51.6	46.8
		MCDB	49.7	48.6	53.7	55.6	63.4	67.7	72.5	72.7	67.1	58.5	53.7	48.3
Mean Daily Temperature Range	5% DB	MDBR	9.6	11.1	12.4	14.0	16.5	16.7	20.0	20.1	16.9	13.8	10.7	9.1
		MCDBR	11.5	14.0	16.0	18.8	23.2	23.3	26.1	26.1	23.5	16.6	12.5	10.4
		MCWBR	7.9	9.6	10.2	11.4	13.3	12.5	12.6	12.3	11.6	10.4	8.9	7.7
	5% WB	MCDBR	10.6	11.7	15.0	17.2	21.2	21.5	24.0	23.5	18.8	13.7	11.5	10.0
		MCWBR	8.4	8.8	10.3	11.3	12.7	12.2	12.7	12.2	10.5	10.0	8.8	8.1
Clear-Sky Solar Irradiance	taub	0.313	0.309	0.324	0.348	0.369	0.353	0.345	0.349	0.335	0.328	0.316	0.310	
	taud	2.536	2.562	2.538	2.460	2.416	2.477	2.526	2.521	2.555	2.568	2.546	2.500	
	Ebn at Noon	244	273	285	287	283	288	289	283	278	261	239	227	
	Edh at Noon	19	23	27	32	35	33	31	30	27	23	19	17	
All-Sky Solar Radiation	RadAvg	297	577	899	1360	1734	1898	2083	1758	1252	690	353	238	
	RadStd	40	76	105	127	137	179	147	114	102	77	54	31	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (68 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page